

LONG POINT, ON, Canada

WMO: 714640

Lat: 42.549N

Lon: 80.049W

Elev: 574

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.1	9.8	-5.4	4.2	4.2	-1.7	5.1	7.4	46.6	26.5	42.1	26.3	20.2	230	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	79.1	70.6	77.5	70.1	76.3	69.3	72.9	76.7	71.8	75.8	70.6	74.9	12.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	119.1	76.0	70.2	113.6	75.1	69.0	108.7	74.2	37.0	76.8	35.9	76.1	34.9	74.8	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.4	35.8	31.7	DB	3.2	83.8	7.0	2.4	-1.8	85.5	-5.8	87.0	-9.7	88.3	-14.8	90.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	27.2	27.1	33.5	43.3	54.5	64.9	71.1	71.2	65.6	54.5	44.1	33.7	(6)	
	DBStd	17.54	9.66	9.73	8.67	7.19	6.97	5.57	4.00	3.77	5.88	7.34	7.98	8.36	(7)	
	HDD50	2873	707	642	516	225	31	0	0	0	0	39	207	506	(8)	
	HDD65	6266	1172	1062	978	652	334	69	5	3	62	333	627	969	(9)	
	CDD50	2637	1	0	4	23	172	447	654	656	470	178	30	2	(10)	
	CDD65	550	0	0	0	0	9	65	193	194	82	7	0	0	(11)	
	CDH74	995	0	0	0	0	12	85	384	390	122	2	0	0	(12)	
	CDH80	26	0	0	0	0	0	3	13	8	2	0	0	0	(13)	
(14) Wind	WSAvg	16.3	20.3	17.9	16.7	16.8	15.0	13.0	12.6	12.7	14.0	17.5	19.5	19.8	(14)	
(15) Precipitation	PrecAvg	37.6	3.2	2.5	2.9	3.3	3.2	3.0	3.4	3.0	3.5	3.3	3.1	3.2	(15)	
	PrecMax	47.1	5.2	5.3	5.5	6.0	6.6	7.7	6.8	5.2	9.2	6.2	5.9	7.0	(16)	
	PrecMin	26.6	1.5	1.4	1.4	1.3	1.3	1.1	0.6	0.8	0.9	1.0	0.7	1.5	(17)	
	PrecStd	5.8	1.0	0.9	1.1	1.1	1.4	1.5	1.4	1.1	1.7	1.2	1.2	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	52.2	60.9	67.8	75.7	79.7	81.5	80.9	79.5	72.6	63.0	56.1	(19)	
		MCWB	47.1	49.7	50.9	58.5	65.4	67.2	70.9	71.9	71.4	64.5	56.9	51.9	(20)	
	2%	DB	47.1	46.5	53.8	61.9	71.0	76.4	79.1	78.8	77.1	69.6	59.8	51.2	(21)	
		MCWB	43.7	43.0	46.6	52.5	63.4	67.2	70.4	71.0	71.2	62.6	54.7	47.3	(22)	
	5%	DB	43.2	42.2	48.4	57.4	67.9	73.9	77.4	77.4	75.2	67.0	57.3	47.8	(23)	
		MCWB	40.0	38.5	44.0	50.3	60.9	66.3	69.9	70.1	69.3	60.7	51.5	45.1	(24)	
	10%	DB	39.5	38.2	44.2	53.2	64.5	71.9	76.1	76.2	73.5	64.4	54.9	44.6	(25)	
		MCWB	36.6	35.2	39.8	46.5	58.1	64.9	69.2	69.2	67.7	58.6	49.3	41.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	50.2	49.0	58.9	67.1	71.3	74.3	74.8	73.8	67.4	59.0	51.8	(27)	
		MCDB	50.9	52.7	52.0	65.6	73.7	75.8	78.0	78.8	77.5	71.7	62.3	55.8	(28)	
	2%	WB	43.5	44.1	45.0	54.6	64.8	69.0	72.7	73.0	72.3	65.5	55.7	47.4	(29)	
		MCDB	46.0	46.8	48.3	60.7	70.0	73.6	76.4	77.0	76.2	69.1	59.8	50.2	(30)	
	5%	WB	39.8	39.5	41.6	50.9	61.8	67.2	71.5	71.7	70.9	63.1	51.8	45.1	(31)	
		MCDB	42.0	42.8	45.4	56.1	66.1	71.3	75.3	75.9	75.1	67.4	56.1	47.8	(32)	
	10%	WB	36.3	35.9	38.2	47.4	58.4	65.4	70.2	70.4	69.2	61.0	49.6	41.3	(33)	
		MCDB	38.6	39.0	41.3	52.5	63.1	69.5	74.4	75.0	73.3	66.2	54.5	44.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.6	9.8	11.3	11.2	8.8	7.1	6.6	7.3	7.8	8.2	7.8	(35)	
		MCDBR	13.0	14.8	16.4	19.0	14.8	10.7	7.8	7.2	7.9	9.3	10.4	10.7	(36)	
		MCWBR	13.3	13.5	13.0	14.0	9.9	7.5	6.0	5.9	6.3	7.7	8.9	11.1	(37)	
	5% WB	MCDBR	13.2	15.8	14.2	18.3	15.0	9.6	7.2	6.1	7.5	8.8	9.7	11.2	(38)	
		MCWBR	12.7	14.5	12.2	15.3	11.5	7.4	5.9	5.9	6.2	8.4	9.2	10.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.301	0.345	0.338	0.380	0.406	0.439	0.434	0.422	0.383	0.361	0.315	0.299	(40)	
	taud		2.417	2.224	2.340	2.311	2.297	2.244	2.282	2.310	2.383	2.419	2.516	2.493	(41)	
	Ebn at Noon		266	267	287	281	276	266	266	266	268	260	261	259	(42)	
	Edh at Noon		24	35	35	39	41	43	41	39	34	29	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		449	665	1038	1424	1762	1947	1944	1713	1344	823	507	310	(44)	
	RadStd		34	77	134	148	172	134	139	91	122	79	75	36	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page